

Nishchal Sapkota

Machine Learning Scientist specializing in vision foundation models, self-supervised and multimodal learning, and healthcare AI

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🏠 [Personal Website](#)

🎓 [Google Scholar](#)

Education

University of Notre Dame (UND)

Notre Dame, IN

Ph.D. in Computer Science and Engineering

12/2026

M.S. in Computer Science and Engineering

08/2024

Research Areas: Computer Vision, Self-supervised Learning, Foundation Models, Data-efficient Deep Learning Models, AI for Healthcare

The University of Southern Mississippi (USM)

Hattiesburg, MS

B.S. in Computer Science | B.S. in Mathematics – Honors, summa cum laude (GPA: 3.91)

08/2020

Thesis: Probabilistic Analysis of Revenues in Online Games

Experiences

Amazon

Sunnyvale, CA

Applied Scientist Intern

08/2026 – 11/2026

- Developing novel reinforcement learning approaches for expert-feedback-driven personalization of vision-language (VLM) and large text-to-image (T2I) models.
- Building large-scale multimodal generation and editing pipelines leveraging diffusion and flow-based models, vision-language reasoning, and controllable image synthesis for fine-grained personalized generation.

Mayo Clinic

Rochester, MN

Computational Pathology and AI Intern

01/2025 – 07/2026

- Developed the **largest pathology vision foundation model** with $300\times$ less data and achieving up to 5% improvement over state-of-the-art methods through scaling-aware training and mid-training regularization to resolve scaling bottlenecks. [1]
- Assisted in extending pathology foundation models into **multimodal vision-language models** by integrating patient diagnosis reports with whole-slide pathology images.
- Built large-scale healthcare AI systems using **distributed training** and scalable **MLOps pipelines** for clinical predictive modeling. [9]

IBM

Durham, NC

Senior Data Science & AI PhD Intern

05/2025 – 08/2025

- Developed **user-behavior and session-level predictive models** (N-gram Markov, Transformer, Mamba) on large-scale web clickstream data to identify high-intent users and predict conversion outcomes, achieving a 14% improvement over internal baselines.
- Deployed models to enable real-time agentic AI marketing interventions, recovering up to **10,000 lost conversion opportunities per quarter**, while collaborating cross-functionally to translate model insights into business strategy.

The University of Notre Dame

Notre Dame, IN

Graduate Researcher

08/2020 – Present

- Designed a memory-augmented **video segmentation model** [2] to advance AI-assisted surgery, outperforming foundational models (SAM2/3) in public surgical benchmarks by up to 17% and developed lightweight variants for real-time deployment.
- Pioneered three data-efficient **3D segmentation models** with up to 11% improvement on medical image/volume segmentation benchmarks. [3][4][5]
- Built **self-supervised learning** frameworks achieving up to 4% gains across medical image segmentation benchmarks. [10][11][12]
- Proposed **multimodal learning** frameworks and shape-aware segmentation/classification models using **implicit neural representations** by handling label ambiguity and improving data efficiency by 30%. [6][13][16]
- Collaborated with academic, clinical, and industry labs** to develop AI-driven healthcare solutions and **mentored 5+ students**, resulting in multiple publications and successful industry and academic placements. [6][14][15][17][18]

The University of Southern Mississippi

Hattiesburg, MS

Undergraduate Researcher

08/2017 – 05/2020

- Analyzed online games using Markov Chain to **maximize revenues** for both players and the providers. [7]
- Introduced a novel **dynamic food chain model** for three species and analyzed its long-term behavior. [8]
- Predicted **chemical compound toxicity** using in-vitro computational methods and feature engineering.

Technical Skills

Programming: Python, R, C++, Bash, MATLAB, SQL

ML Packages: PyTorch, Tensorflow, wandb, HuggingFace

Tools: Jupyter, LaTeX, FIJI, Microsoft 365, Adobe Illustrator, Docker

Concepts: Machine Learning, Computer Vision, Foundation Models, Self-supervised Learning, Multimodal Learning, Temporal Modeling for Video, Video Understanding, 3D Segmentation, Low-latency Inference, Diffusion, Latent Flow Matching, Distributed Training, Model Quantization, Domain Adaptation, Computational Pathology, Medical Image Analysis, Time Series Forecasting, Mathematical Modeling

Math Concepts: Data Analysis, Numerical Methods, Real Analysis, Modern Algebra, Number Theory, Statistics, Probability

Publications

First Author

[1] [BM1B: Data-Efficient Scaling of Bone Marrow Foundation Models through Domain-Specific Self-Supervised Learning and Dense Morphology-Preserving Representation Learning](#) – *MedFMB @ ECCV*, 2026

[2] [Sparsely Supervised Surgical Video Segmentation with Reliable Asymmetric Dual Memory](#) – *MICCAI*, 2026

[3] [When Swin Transformer Meets KANs: An Improved Transformer Architecture for Medical Image Segmentation](#) – *IEEE ISBI*, 2026

[4] [Universal Conditional Networks for Cartilage Segmentation with Sparsely Annotated Data](#) – *Nature Scientific Reports*, 2025

[5] [ConUNETR: A Conditional Transformer Network for 3D Micro-CT Embryonic Cartilage Segmentation](#) – *IEEE ISBI*, 2024

[6] [SHMC-Net: A Mask-guided Feature Fusion Network for Sperm Head Morphology Classification](#) – *IEEE ISBI*, 2024

[7] [Probabilistic Analysis of Revenues in Online Games](#) – *University of Southern Mississippi*, 2020

[8] [Hunting Co-operation in the Middle Predator in Three Species Food Chain Model](#) – *MAA Proceedings*, 2020

Co-authored

[9] [Lymphovision: A lymphoma-specialized foundation model for histology-based lymphoma classification and subtyping](#) – *Blood*, 2025

[10] [A Point in the Right Direction: Vector Prediction for Spatially-aware Self-supervised Volumetric Rep. Learning](#) – *IEEE ISBI*, 2023

[11] [Keep Your Friends Close & Enemies Farther: Debiasing Contrastive Learning with Spatial Priors in 3D Images](#) – *IEEE BIBM*, 2022

[12] [Unsupervised Feature Clustering Improves Contrastive Representation Learning for Medical Image Segmentation](#) – *IEEE BIBM*, 2022

[13] [SwIPE: Efficient and Robust Medical Image Segmentation with Implicit Patch Embeddings](#) – *MICCAI*, 2023

[14] [H-CNN-ViT: A Hierarchical Gated Attention Multi-Branch Model for Bladder Cancer Recurrence Prediction](#) – *IEEE BIBM*, 2025

[15] [IHCSurv: Effective Immunohistochemistry Priors for Cancer Survival Analysis in Gigapixel Multi-stain WSIs](#) – *MICCAI*, 2024

[16] [Path-GPTOmic: A Balanced Multi-modal Learning Framework for Survival Outcome Prediction](#) – *IEEE ISBI*, 2024

[17] [Boosting Medical Image Classification with Segmentation Foundation Model](#) – *IEEE ISBI*, 2024

[18] [Embryonic Cranial Cartilage Defects in the Fgfr^{3Y367C/+} Mouse Model of Achondroplasia](#) – *Anatomical Record*, 2023

Scholarships, Grants, Honors, and Achievements

2024 IEEE International Symposium on Biomedical Imaging (ISBI2024) Travel grant (\$800)	ISBI 2024
Graduate School Professional Development Fund (\$1, 250) and Conference Presentation Grant (\$450)	UND 2024
CSE Select Fellowship Award (1/40 incoming Ph.D students; yearly stipend worth \$40, 000)	UND 2020-2025
Wright W. and Annie R. Cross Endowment (\$10, 500) and Danny R. Carter Endowed Scholarship (\$4, 000)	USM 2017-2020
Eagle SPUR grant for Undergraduate Research (\$2, 000) and Honors Keystone Scholarship (\$2, 000)	USM 2019
First Place , Mathematics Comprehensive Exam (MFT)	USM 2019
Second Runner Up : Best Undergraduate Paper	MAA Meeting 2019
Finalist , Integration Bee	MAA Meeting 2018
Nominated for College of Science and Technology's Outstanding Sophomore Award	USM 2017
Burner Science & Tech. Scholarship (\$800), Wallace C. & Lynn L. Pye Endowed Scholarship (\$800)	USM 2017

Teaching Experiences

The University of Notre Dame	Notre Dame, IN
Graduate Teaching Assistant	08/2020 – 12/2025
• Design & Analysis of Algorithms; Complexity & Algorithms; Mobile App. Design; Discrete Mathematics • Prepared lecture slides, graded submissions, created answer keys, and held office hours.	
STEM Project Leader Warrior-Scholar Project	06/2023, 06/2024
• Medical Image Analysis: Designed and conducted a Bootcamp to prepare veterans for undergraduate research. • Introduction to Data Science: Conducted a Bootcamp to prepare veterans for undergraduate coding classes.	

Students Mentored

- **Haoyan Shi** (Undergraduate Intern, 2025) – *Published author, Currently a PhD student at University of Notre Dame*
- **Zihao Zhao** (Undergraduate Intern, 2024) – *Currently a PhD student at Rutgers*
- **Maria Jose Gomez** (High School Intern, 2024) – *Published author, applying to undergraduate programs*
- **Sirui Li** (Undergraduate Intern, 2023) – *Currently a PhD student at UCLA*
- **Santiago Rodriguez** (Undergraduate Intern, 2023) – *Now a Software Engineer at Apple*